Table 7. Quantitative dewarping performance comparisons on the DIR300 benchmark dataset.

Method	MS-SSIM↑	LD ↓	AD ↓	CER ↓	ED ↓	Para.
Deformed	0.317	39.58	0.771	0.372	1434.54	-
DewarpNet [Das et al. 2019]	0.492	13.94	0.332	0.281	1058.36	86.9M
DocTr [Feng et al. 2021]	0.616	7.23	0.256	**0.173**	589.98	26.9M
DDCP [Xie et al. 2021]	0.552	10.97	0.356	0.554	2189.93	13.3M
PaperEdge [Ma et al. 2022]	0.584	7.96	0.254	0.192	594.63	36.6M
DocGeoNet [Feng et al. 2022]	0.638	6.40	0.242	0.219	664.95	24.8M
FTA [Li et al. 2023b]	0.607	7.68	0.244	0.184	668.95	45.2M
UVDoc [Verhoeven et al. 2023]	0.621	7.72	0.218	0.203	608.47	8M
LADoc [Li et al. 2023a]	0.652	5.70	0.195	0.192	546.32	-
DvD	**0.680**	**5.01**	**0.173**	0.188	**528.52**	151.25M

Table 8. Model architecture settings on CEB and FGB.

Setting	Value
CEB: Condition Embedding Blocks	
Block layers	12
Conditioning mechanism	cross-attention
Cross Attention query	noisy latent variable m_T
Cross Attention key, value	composite condition c_t
CEB base architecture	DiT_S_2
Hidden dimension	384
Number of attention head	6
Image Input dimension	512×512×3
Transformer patch size	2
Latent space size	64×64
FGB: Fusion Generation Blocks	
Block layers	6
Fusion mechanism	self-attention
Hidden dimension	1536
Number of attention head	6
Feed forward layers	MLP
Output dimension	64×64×2

A TRAINING AND EVALUATION SETTINGS

During the training phase, we develop the proposed DvD using PyTorch 2.0.1. The model is trained for 1,800,000 iteration with a batch size of 10 on four NVIDIA GeForce RTX 3090 GPUs. The learning rate is set to 1E-4 with Adam optimizer. During the evaluation stage, to ensure a fair comparison, we measure the dewarped results of previous work in a strictly unified software environment. Specifically, following the Ma et al. [Ma et al. 2018], we use MATLAB 2022A to measure feature alignment metrics (i.e., MS-SSIM, LD, AD). For OCR metrics, we use the unified Tesseract OCR engine v5.1.0[1]. For MLLM-based OCR metrics, we use the open-source QWen2.5-VL 7B [Bai et al. 2025] as the OCR engine. We also find that the MLLMs have a issue of being prone to language repeater, which means MLLMs continuously produce a series of repetitive tokens. To address this issue, we set rules to detect and delete these redundant tokens. During the ablation stage, we uniformly apply the most widely used DocUNet [Ma et al. 2018] benchmark to present component analysis.

[1]https://github.com/UB-Mannheim/tesseract/commits/v5.1.0.20220510

Table 9. Quantitative dewarping performance comparisons on the DocReal benchmark dataset.

Method	MS-SSIM↑	LD ↓	AD ↓	CER ↓	ED ↓	Para.
Deformed	0.320	35.81	0.980	0.6552	676.35	-
DewarpNet [Das et al. 2019]	0.484	13.54	0.338	0.480	478.82	86.9M
DocTr [Feng et al. 2021]	0.555	12.71	0.324	0.467	458.13	26.9M
DDCP [Xie et al. 2021]	0.473	16.33	0.451	0.583	622.46	13.3M
PaperEdge [Ma et al. 2022]	0.523	11.34	0.297	0.417	403.46	36.6M
DocGeoNet [Feng et al. 2022]	0.557	12.12	0.316	0.468	447.08	24.8M
FTA [Li et al. 2023b]	0.517	13.24	0.347	0.457	449.37	45.2M
UVDoc [Verhoeven et al. 2023]	0.554	11.32	0.271	0.445	430.40	8M
Docreal [Yu et al. 2024]	0.559	9.83	0.282	0.458	414.91	-
DvD	0.564	11.69	0.277	0.446	427.48	151.25M

Table 10. Inference time comparison (seconds).

Methods	**Inferring time (s) ↓**
DewarpNet [Das et al. 2019]	1.61
DocTr [Feng et al. 2021]	0.54
DDCP [Xie et al. 2021]	0.71
DocGeoNet [Feng et al. 2022]	0.44
FTA [Li et al. 2023b]	0.82
DvD	0.59

B MORE NETWORK ARCHITECTURE SETTINGS

In this section, we present a more detailed architecture of our diffusion decoder, which consists of 12 Condition Embedding Blocks (CEB) and 6 Fusion Generation Blocks (FGB), as shown in Tab. 8. In each CEB, we focus on building the conditional embedding layer for our conditional diffusion model by extending the standard diffusion transformer [Peebles and Xie 2023] with a cross-attention mechanism. Concretely, we adopt the typical *DiT_S_2* as base block architecture. Before feeding inputs into the CEB, we downsample all compound conditions c_t to a 64×64 latent space and project them into the 384-dimensional DiT embedding via a patch-embedding layer. Within each CEB, the noised mapping variable m_t serves as the transformer's query, while the four components of the composite condition c_t serve as the keys and values. All four conditions are embedded in parallel using shared parameters, and their outputs are concatenated to form a unified conditional embedding. This embedding is then fed to the FGB. In each FGB, we apply four self-attention layers along with feed-forward layers over these concatenated features to achieve feature fusion. Finally, a stack of three linear layers along with an unpatchify module is employed to yield the final 2D mapping for dewarping.

C MORE EXPERIMENTS ON DIR300, DOCREAL AND ANYPHOTODOC6300

In this section, we provide supplementary quantitative comparison experiments to those presented in the main paper, focusing on the evaluation on DIR300 [Feng et al. 2022], DocReal [Yu et al. 2024], and the overall quantitative evaluation on our AnyPhotoDoc6300, as shown in Tab. 7, Tab. 9, and Tab. 12, respectively. It is worth

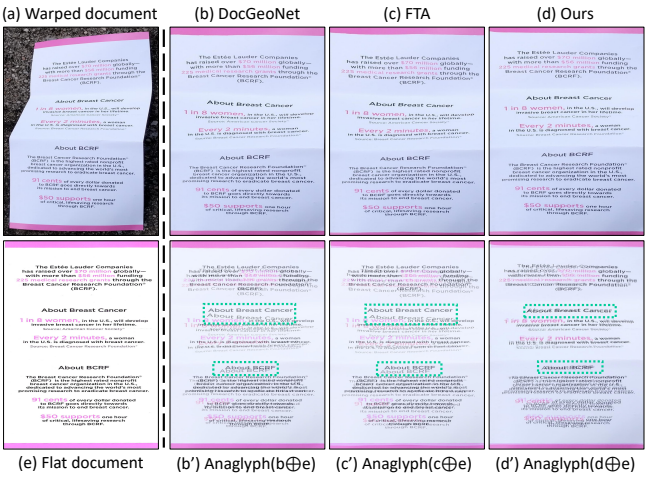


Fig. 13. **Document content structure preservation** (a) is an input warped document. (b~d) is dewarped results from (b) DocGeoNet [Feng et al. 2022], (c) FTA [Li et al. 2023b], (d) Our DvD. (e) is a ground truth flat document for reference. For better comparison, we provide anaglyphs (b'~d') that illustrate the large spatial misalignment between dewarped (b~d) and flat document (e), denoted as "⊕". We highlight some conspicuously misaligned regions by dotted lines. Compared with previous state-of-the-art, our DvD performs less content misalignment.

noting that some models either do not open their inference code or provide the dewarped results on DIR300 and AnyPhotoDoc6300; therefore, we do not account for them. From the results, we can get a similar conclusion to that from the DocUNet benchmark listed in the main paper. In these tables, we overall achieve competitive performance on most metrics, most of which we perform the best or second-best. Particularly, in Tab. 12, we achieve the top scores on all metrics except LD and MMCER. These results demonstrate that DvD is highly competitive both in broad and fine-grained evaluations.

Table 11. **Statistics of AnyPhotoDoc6300.** Flat Source: flat source document obtained by scanner; Photo Doc.: Number of Photographic Document Images; WP: Warping Pattern. Domains: LC (Layout Category), EL (Environment Lighting), CA (Capture Angle)

LC	Source Flat	WP	EL	CA	Photo Doc.
Single-column Layout Paper	50	×3	×3	×2	900
Double-column Layout Paper	50	×3	×3	×2	900
Bound Book	100	×1	×3	×2	600
Complex Layout Paper	50	×3	×3	×2	900
Bill	51	×3	×3	×2	306
Invoice	50	×3	×3	×2	900
Magazine	50	×3	×3	×2	900
Education Script	50	×3	×3	×2	900
Total					6306

D AN VISUALIZATION EXPLANATION FOR STRUCTURE PRESERVATION

Relying solely on a single warped image as input to estimate the true positions of text lines places heavy constraints on the process, leading to imprecise structure preservation. For clarity, Fig. 13 shows a visualization example. Although current dewarping models from (b) and (c) produce plausible dewarped outputs, they cannot be aligned exactly with the true ground truth in (e), which limits the possibility of accurately restoring the document to its scanned appearance. In the second row, we overlay the ground truth image with each dewarped result. As can be seen, DvD substantially alleviates the spatial misalignment for document content, thereby offering better structure preservation.

E MORE COLLECTION DETAILS AND VISUALIZATION ABOUT ANYPHOTODOC6300 BENCHMARK

AnyPhotoDoc6300 enables fine-grained document dewarping evaluation under three warping patterns across three distinct domains(i.e., Layout Categories, Environment Lighting, Capture Angle). In this section, we provide supplementary collection details and more visualization on the next page. The basic composition and statistics of our dataset are summarized in Table 11. We begin by crawling a source collection of document images from the Internet and manually screening all low-resolution images (less than 1000×1000 px). In such a way, we collect the set of *flat images* for each category except the Bound Book and Bill categories. Every category contains exactly 50 flat images. Besides, the Bound Book category has 100 images because we collect both 50 single-page scans and 50 double-page scans as its *flat images*. The Bill category contains 51 images due to its limited availability on the Internet. For the Bill category, to achieve a similar evaluation, we quantitatively divided 17×3 bill documents, where the 3 indicates that multiple bills are divided into three groups according to the warping pattern. It's worth noting that due to the issue of the automatic margins mechanism in the printed documents, we adopted the following strategy to ensure the precise alignment of the photographed and the scanned documents. Specifically, we print each source *flat images* on standard A4 paper using a high-quality printer Ricoh MC2000ew. We then scan all these A4 prints, as well as real books and bills, on a flatbed scanner. These scanned results constitute the *flat images* for our benchmark.

Table 12. Quantitative dewarping performance comparisons on the AnyPhotoDoc6600 benchmark dataset. **Bold** indicates the best, underline indicates second-best. The last column shows the network size by the number of parameters (millions).

Method	Venue	Training Dataset	MS-SSIM ↑	LD ↓	AD ↓	CER ↓	ED ↓	MMCER ↓	MMED ↓	Para.
Warped Document	-	-	0.244	37.85	1.141	0.845	1138.99	0.561	770.022	-
DDCP [Xie et al. 2021]	ICDAR'21	DDCP	0.429	16.61	0.482	0.767	957.59	0.763	1464.65	13.3M
PaperEdge [Ma et al. 2022]	SIGGRAPH'22	Doc3D+DIW	0.463	**14.07**	0.403	0.701	761.43	0.470	887.61	36.6M
UVDoc [Verhoeven et al. 2023]	SIGGRAPHA'23	Doc3D+UVDoc	0.482	14.83	0.380	0.722	793.63	0.691	636.37	8M
DewarpNet [Das et al. 2019]	ICCV'19	Doc3D	0.423	15.54	0.453	0.743	879.51	0.433	594.40	86.9M
DocTr [Feng et al. 2021]	MM'21	Doc3D	0.477	15.82	0.410	0.710	806.82	**0.284**	412.08	26.9M
DocGeoNet [Feng et al. 2022]	ECCV'22	Doc3D	0.486	15.08	0.407	0.714	819.69	0.303	434.70	24.8M
FTA [Li et al. 2023b]	ICCV'23	Doc3D	0.470	15.79	0.426	0.720	832.30	0.314	438.23	45.2M
DvD	-	Doc3D	**0.504**	14.47	**0.357**	**0.700**	**766.31**	0.312	**394.68**	151.25M

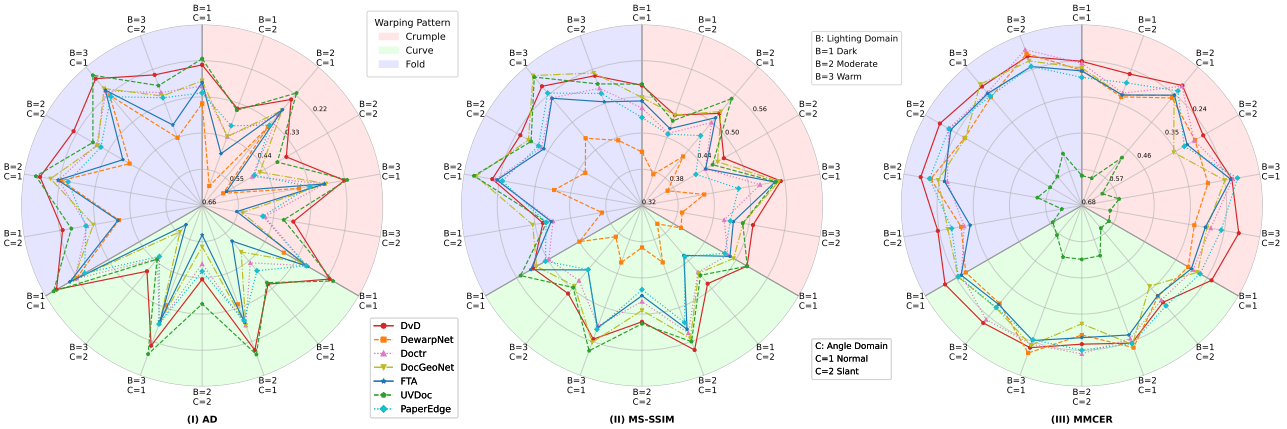


Fig. 14. More Quantitative dewarping performance comparisons on the AnyPhotoDoc6600 benchmark dataset. Under the fixed "Complex Layout Paper" Layout Pattern, we provided 18 dimensions of evaluation for AD, MS-SSIM, and MMCER.

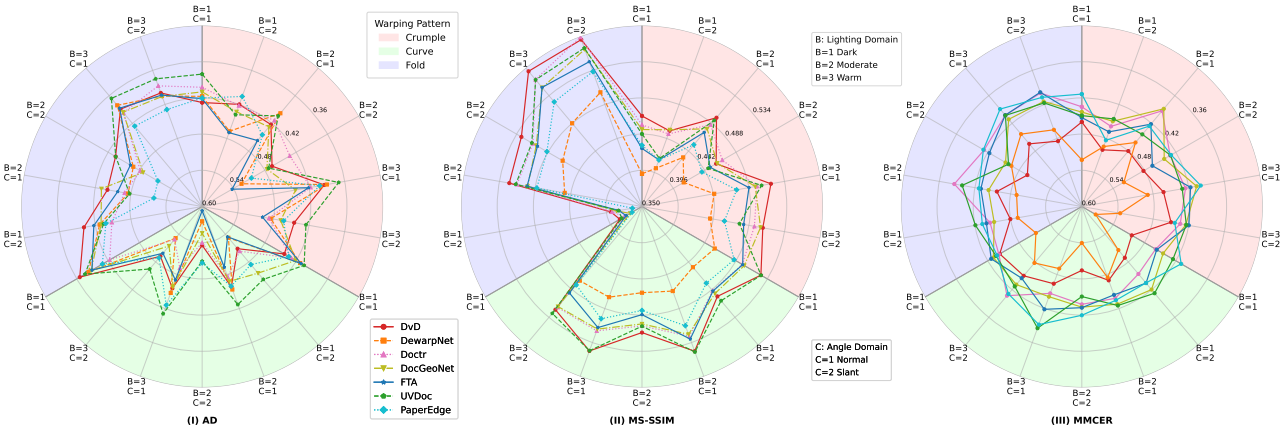


Fig. 15. More Quantitative dewarping performance comparisons on the AnyPhotoDoc6600 benchmark dataset. Under the fixed "Invoice" Layout Pattern, we provided 18 dimensions of evaluation for AD, MS-SSIM, and MMCER.

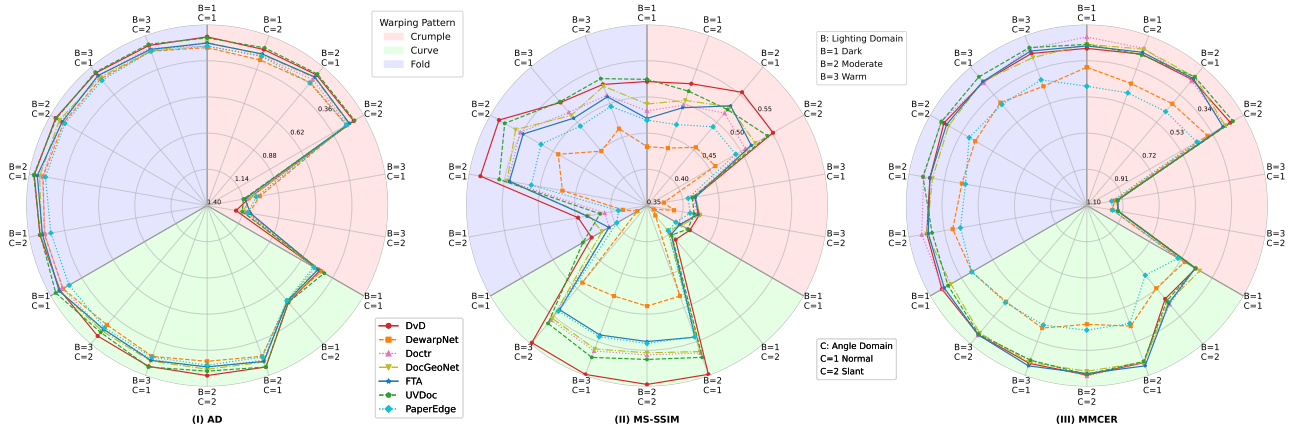


Fig. 16. More Quantitative dewarping performance comparisons on the AnyPhotoDoc6600 benchmark dataset. Under the fixed "Education Script" Layout Pattern, we provided 18 dimensions of evaluation for AD, MS-SSIM, and MMCER.

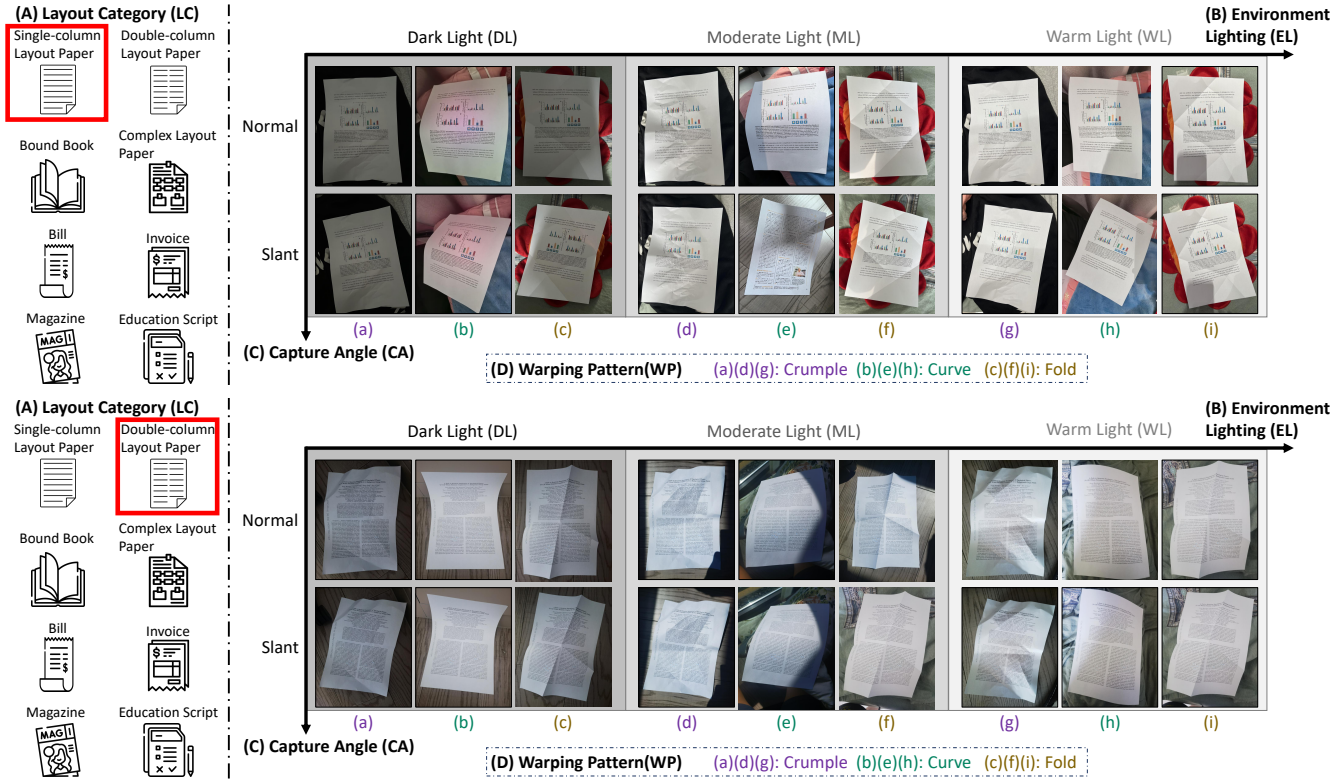


Fig. 17. More sample data array visualization for "Single-column Layout Paper" and "Double-column Layout Paper" in our AnyPhotoDoc6300 Benchmark. We aim to evaluate the dewarping capability of the model to different warping patterns (D) under well-distinguished domain combinations (A, B, C).

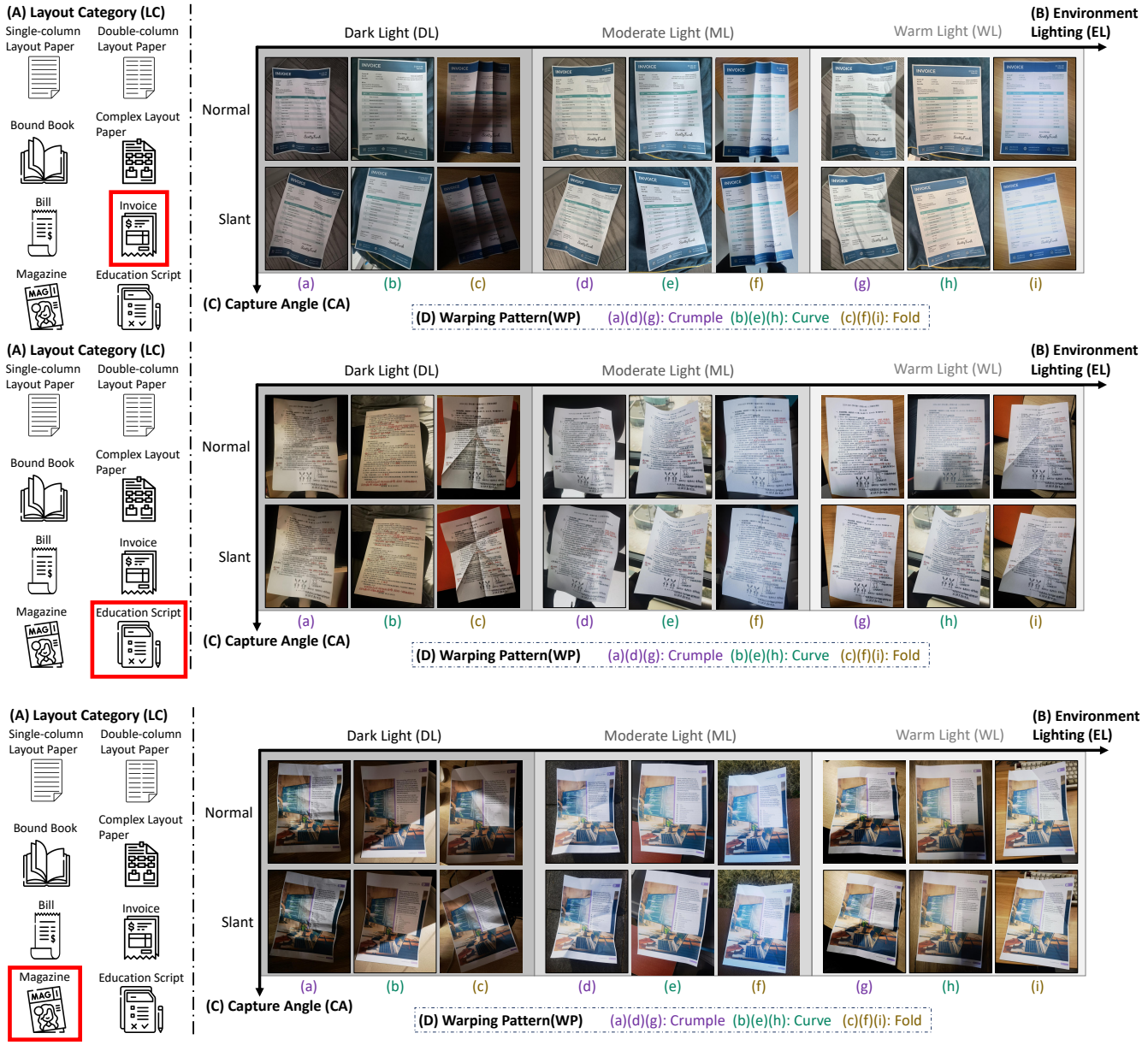


Fig. 18. More sample data array visualization for "Invoice", "Education Script", and "Magazine" in our AnyPhotoDoc6300 Benchmark. We aim to evaluate the dewarping capability of the model to different warping patterns (D) under well-distinguished domain combinations (A, B, C).

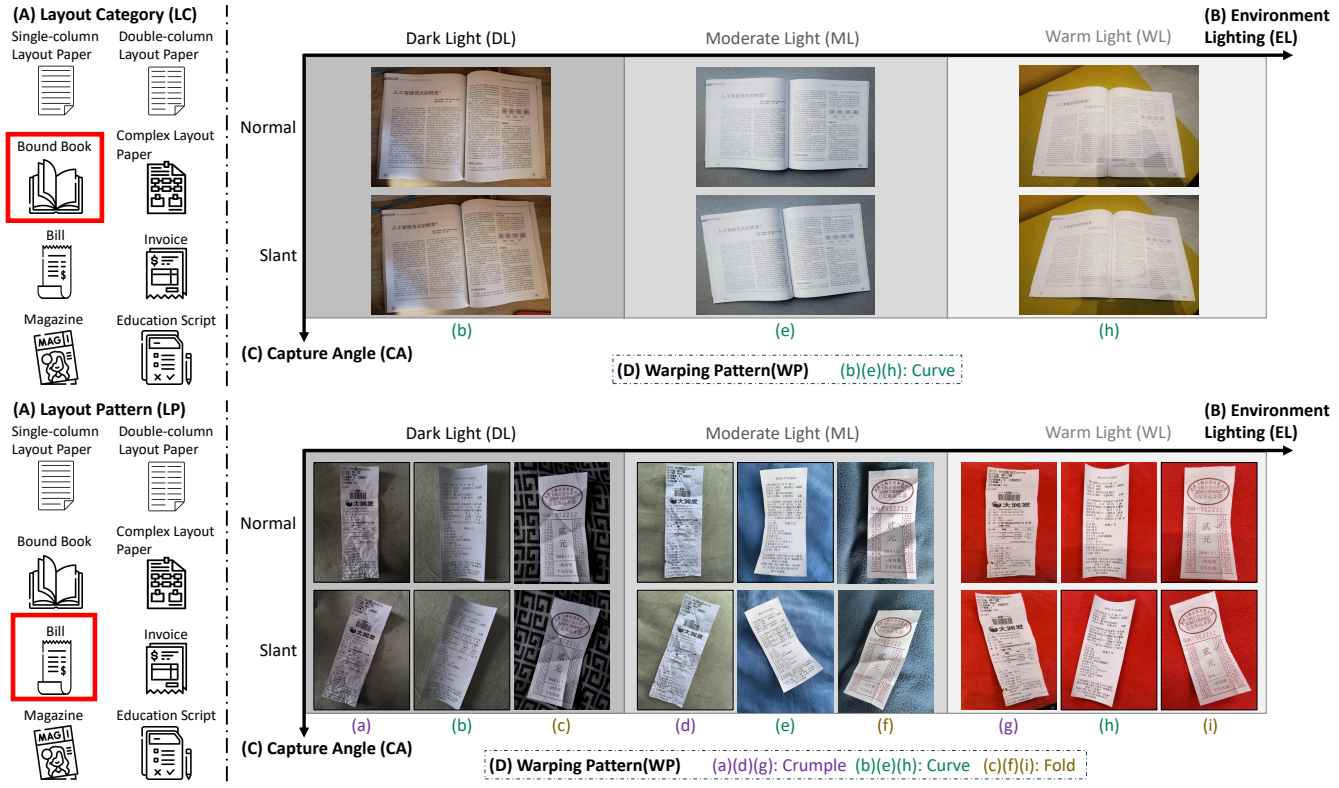


Fig. 19. More sample data array visualization for "Bound Book" and "Bill" in our AnyPhotoDoc6300 Benchmark. We aim to evaluate the dewarping capability of the model to different warping patterns (D) under well-distinguished domain combinations (A, B, C).